

Installing Docker (CLI & Desktop)

Covers setup for:

- Ubuntu Linux
- macOS
- Windows (WSL2)

1. Docker on Ubuntu Linux

Install Docker Engine (CLI)

```
sudo apt update
sudo apt install -y \
    ca-certificates \
    curl \
    gnupg \
    lsb-release
```

Add Docker's GPG key:

```
sudo mkdir -p /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg \
| sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
```

Ubuntu: Add Docker Repo & Install

```
echo \
  "deb [arch=$(dpkg --print-architecture) \
  signed-by=/etc/apt/keyrings/docker.gpg] \
  https://download.docker.com/linux/ubuntu \
  $(lsb_release -cs) stable" | \
  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

sudo apt update
sudo apt install -y docker-ce docker-ce-cli containerd.io
```

Enable & start Docker:

```
sudo systemctl enable docker
sudo systemctl start docker
```

Optional: Install Docker Desktop (GUI)

1. Download `.deb` from:

👉 <https://www.docker.com/products/docker-desktop/>

2. Install it:

```
sudo apt install ./docker-desktop-<version>.deb
```

3. Launch from app menu or `docker-desktop`

2. Docker on macOS

Install Docker Desktop (includes CLI + GUI)

1. Go to:

 <https://www.docker.com/products/docker-desktop/>

2. Download the `.dmg` file for macOS

3. Install and drag Docker to Applications

4. Run Docker Desktop app

 CLI tools like `docker` and `docker-compose` will now work in Terminal

macOS (Optional): CLI only

Use Homebrew:

```
brew install docker
```

You'll still need **Docker Desktop** to run containers (for Docker Engine)

3. Docker on Windows (WSL2)

Install WSL2

```
wsl --install
```

Reboot, then install Ubuntu from Microsoft Store.

Install Docker Desktop (Windows)

1. Download from:

 <https://www.docker.com/products/docker-desktop/>

2. Install Docker Desktop

3. Enable WSL2 backend during setup

4. Start Docker Desktop

 Works inside WSL terminal (`wsl`)

Verify Docker Installation

Check version:

```
docker --version
```

Test:

```
docker run hello-world
```

If you see a greeting from Docker — success! 🎉

Recap

OS	CLI Install	GUI Install
Ubuntu	apt + GPG	Docker Desktop <code>.deb</code>
macOS	Homebrew or Docker Desktop	Docker Desktop (recommended)
Windows	via WSL2	Docker Desktop (with WSL2 backend)

Tips

- Add yourself to the `docker` group on Linux:

```
sudo usermod -aG docker $USER  
newgrp docker
```

- Use Docker Desktop for Kubernetes support & GUI dashboards
- Restart your terminal after installation

 **You're Ready to Containerize!**